

Sanket Kumar

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SUMMARY

Computer Science undergraduate with hands-on experience in Backend Development, DevOps, and Artificial Intelligence through academic projects. Strong foundation in Java, Python, REST APIs, and automation, with an interest in building scalable applications.

EDUCATION

Vellore Institute of Technology	2023 – Present
<i>Bachelor of Technology in Computer Science CGPA: 8.69</i>	<i>Vellore</i>
Baldwin Farm Area High School	2023
<i>Class XII 80%</i>	<i>Jamshedpur</i>
Baldwin Farm Area High School	2021
<i>Class X 93%</i>	<i>Jamshedpur</i>

PROJECTS

Fruit Quality Detection System | *Python, TensorFlow Lite, OpenCV, Raspberry Pi*

- Designed and deployed a real-time fruit classification system on Raspberry Pi using TensorFlow Lite and OpenCV.
- Integrated an MQ-135 ethylene gas sensor with ADS1115 ADC to classify fruit ripeness as *Fresh*, *Ripening*, or *Overripe*.
- Enabled early detection of spoiled fruits to improve food quality monitoring and reduce supply chain waste.

DevOps-Enabled To-Do REST API | *Java, Jenkins, Docker, Terraform, Ansible*

- Built and tested a RESTful Java API using JUnit with an automated CI/CD pipeline powered by Jenkins.
- Containerized the application with Docker and provisioned infrastructure using Terraform and Ansible.
- Monitored application performance and system health using Graphite and Grafana dashboards.

Adaptive Cyber Defense System (Reinforcement Learning) | *Python, PyTorch, NumPy*

- Developed a multi-adversary cyber defense system using Deep Q-Networks (DQN).
- Simulated evolving attacker behavior and adaptive defense actions (Allow, Block, Throttle, Observe, Deceive) using reinforcement learning.
- Trained the model over 30,000 simulation steps to achieve stable policy convergence.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, HTML/CSS

Technologies: Docker, Jenkins, Terraform, Ansible, Git, GitHub, React.js, Node.js, Django, Angular, TensorFlow, PyTorch, OpenCV, Raspberry Pi, JUnit

Databases: Oracle SQL, MySQL, Firebase

Concepts: Data Structures & Algorithms, Object-Oriented Programming, Compiler Design, Operating Systems, REST API Development, Artificial Intelligence, DevOps, Cloud Computing

CERTIFICATIONS

- **DevOps, Agile & Design Thinking** – IBM Career Education Program *Jul 2026*
Credential ID: CEDEA1IN — Certificate
Hands-on exposure to DevOps practices, Agile workflows, Design Thinking, and CI/CD pipelines.
- **DevOps Fundamentals** – IBM Career Education Program *Jun 2025*
Credential ID: CEDEV2IN — Certificate
Covered Docker, automation, cloud deployment, infrastructure provisioning, and CI/CD fundamentals.

LEADERSHIP & EXTRACURRICULARS

- **Senior Core Member, Geospatial Club – VIT** *Technical Domain*
Developed and maintained the club website while organizing technical workshops and events.